

Shock Wave Dynamics in Traffic Flow: LWR Model Calibration and Validation with NGSIM Data

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MA 8213: Foundations of Applied Mathematics II

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Spring 2026

Abstract

This paper investigates the ability of first-order nonlinear conservation laws to capture the formation and propagation of traffic congestion waves in real highway data. The Lighthill–Whitham–Richards (LWR) model is derived from first principles and calibrated using high-resolution vehicle trajectory data from the NGSIM US-101 dataset. Empirical fundamental diagrams are constructed using Edie’s definitions, and the Greenshields model is fitted to obtain macroscopic traffic parameters.

Analytical results from the method of characteristics are used to predict shock formation and propagation, and the Rankine–Hugoniot condition is applied to estimate shock speeds. These predictions are compared with observed congestion waves extracted from space–time speed contours. Despite its simplicity, the LWR model reproduces the qualitative structure of backward-propagating traffic shocks and provides reasonable quantitative agreement in shock speeds.

A Godunov finite volume scheme is implemented to solve the model numerically, and simulation results are compared with both analytical solutions and empirical observations. The results highlight both the strengths and limitations of first-order traffic flow models and provide a foundation for incorporating data-driven corrections in future work.

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1 Introduction and Motivation

Traffic congestion is something that affects nearly every major city in the world. Drivers in large urban areas spend dozens of extra hours each year stuck in traffic, and the economic cost of this wasted time and fuel runs into billions of dollars annually. But beyond its practical importance, traffic flow turns out to be a surprisingly rich area for applied mathematics.

The key insight that motivates this project is that traffic jams behave like *shock waves*. The same kind of nonlinear conservation laws that govern shock formation in gas dynamics and fluid mechanics also describe how congestion forms and propagates along a highway. When a region of high-density traffic meets a region of lower density, a sharp transition can develop and move backward through the traffic stream, much like a pressure shock in a gas. Drivers experience this as the familiar stop-and-go pattern: you are cruising along, and then suddenly you have to brake hard, only to speed up again a few minutes later with no obvious cause for the slowdown.

This connection between traffic flow and wave theory was first established in the 1950s by Lighthill and Whitham [2] and independently by Richards [3]. Their model, now known as the Lighthill-Whitham-Richards (LWR) model, treats traffic as a compressible fluid and derives a first-order hyperbolic PDE from a conservation law for vehicle density. Despite being over 60 years old, the LWR model remains one of the most widely used macroscopic traffic flow models, and it continues to appear in modern research on traffic state estimation [8], heterogeneous traffic modeling [9], and numerical methods for traffic networks [10, 11].

1.1 Objectives of This Project

The goal of this project is to carry out a complete mathematical analysis of the LWR traffic flow model, from the derivation of the governing equation all the way to numerical simulation and comparison with real data. Specifically, I will:

- (i) Derive the LWR conservation law from first principles and explain the role of the fundamental diagram.
- (ii) Use the method of characteristics to analyze how initial density profiles evolve over time, and determine when and where shock waves form.
- (iii) Apply the Rankine-Hugoniot jump condition to compute the speed of traffic shocks.
- (iv) Study Burgers' equation as a model for the internal structure of the shock, where nonlinear advection is balanced by diffusion.

- (v) Implement a Godunov-type finite volume scheme to solve the LWR equation numerically.
- (vi) Calibrate and validate the model using real vehicle trajectory data from the NGSIM dataset [5], published by the Federal Highway Administration.

1.2 Outline of the Report

The rest of this report is organized as follows. Section 2 derives the LWR conservation law and introduces the fundamental diagram, including the Greenshields speed-density model. Section 3 discusses the empirical calibration of the fundamental diagram from NGSIM data. Section 4 develops the method of characteristics and analyzes wave breaking. Section 5 covers shock waves, the Rankine-Hugoniot condition, and entropy criteria for selecting admissible solutions. Section 6 solves several Riemann problems that model typical traffic scenarios. Section 7 examines Burgers' equation and the viscous shock structure. Section 8 describes the Godunov numerical scheme. Section 9 presents the simulation results and comparisons with real data. Section 10 discusses the main limitations of the model. Finally, Section 11 gives the conclusions.

2 Derivation of the LWR Conservation Law

In this section, I derive the governing equation for the LWR traffic flow model. The derivation follows the same approach used for conservation laws in fluid mechanics and is based on Chapter 7 of Logan [1], adapted to the specific context of vehicular traffic.

2.1 The Conservation Principle

Consider a single-lane highway that we model as a one-dimensional domain along the x -axis. Let $\rho(x, t)$ denote the *traffic density*, defined as the number of vehicles per unit length at position x and time t . The units of ρ are typically vehicles per mile (veh/mi) or vehicles per kilometer (veh/km).

Now consider a fixed segment of the highway between two points $x = a$ and $x = b$. The total number of vehicles in this segment at time t is

$$N(t) = \int_a^b \rho(x, t) dx. \tag{1}$$

Since vehicles are neither created nor destroyed on this stretch of highway (we assume no

on-ramps or off-ramps for now), the only way $N(t)$ can change is through vehicles entering or leaving at the endpoints. If we define the *traffic flux* $q(x, t)$ as the number of vehicles passing a point x per unit time (with units veh/hr), then the rate of change of $N(t)$ is

$$\frac{dN}{dt} = q(a, t) - q(b, t), \quad (2)$$

where $q(a, t)$ is the inflow at the left boundary and $q(b, t)$ is the outflow at the right boundary. Writing this out using Leibniz's rule gives

$$\int_a^b \frac{\partial \rho}{\partial t} dx = q(a, t) - q(b, t) = - \int_a^b \frac{\partial q}{\partial x} dx. \quad (3)$$

Since this must hold for every interval $[a, b]$, and assuming ρ and q are smooth enough, we can bring everything under one integral and conclude that the integrand must be zero:

$$\frac{\partial \rho}{\partial t} + \frac{\partial q}{\partial x} = 0. \quad (4)$$

This is the *conservation law* for traffic flow. It simply says that vehicles are conserved: any change in density at a point must be accounted for by a change in the flux through that point. Equation (4) is the same form as the conservation laws that appear in fluid dynamics, gas dynamics, and many other areas of continuum mechanics (see Section 6.2 of Logan [1]).

2.2 The Fundamental Diagram

Equation (4) has two unknowns, ρ and q , so we need one more relation to close the system. In the LWR model, we assume that there is a functional relationship between the flux and the density:

$$q = q(\rho). \quad (5)$$

This is an *equilibrium assumption*: it says that drivers adjust their speed instantaneously based on the local density around them. While this is clearly an idealization, it is a reasonable approximation for traffic on a long stretch of highway where conditions change slowly.

Since the flux is the product of density and speed, $q = \rho v$, the assumption (5) is equivalent to saying that the speed depends only on the density:

$$v = v(\rho). \quad (6)$$

The relationship $q(\rho)$ is called the *fundamental diagram* of traffic flow. It is one of the most important concepts in traffic flow theory, and it has been the subject of empirical study since

the pioneering work of Greenshields in 1935 [4].

Some basic physical requirements constrain the shape of $v(\rho)$ and $q(\rho)$:

- When the road is empty ($\rho = 0$), drivers travel at some maximum speed $v_{\max}$, called the *free-flow speed*. So $v(0) = v_{\max}$.
- When the road is completely jammed ($\rho = \rho_{\max}$, the *jam density*), vehicles cannot move. So $v(\rho_{\max}) = 0$.
- Speed should decrease as density increases: $v'(\rho) < 0$ for $0 < \rho < \rho_{\max}$.
- The flux is zero at both extremes: $q(0) = 0$ (no cars, no flow) and $q(\rho_{\max}) = 0$ (complete jam, no flow). In between, $q(\rho)$ has a maximum at some critical density ρ_c .

2.3 The Greenshields Model

The simplest model that satisfies all of these requirements is the linear speed-density relationship proposed by Greenshields [4]:

$$v(\rho) = v_{\max} \left(1 - \frac{\rho}{\rho_{\max}} \right). \quad (7)$$

This says that speed decreases linearly from $v_{\max}$ when the road is empty to zero when the road is at jam density. The corresponding flux function is

$$q(\rho) = \rho v(\rho) = v_{\max} \rho \left(1 - \frac{\rho}{\rho_{\max}} \right). \quad (8)$$

This is a concave, downward-opening parabola in ρ , with $q(0) = q(\rho_{\max}) = 0$ and a maximum at the critical density $\rho_c = \rho_{\max}/2$. At this density, the flux reaches its maximum value

$$q_{\max} = \frac{v_{\max} \rho_{\max}}{4}. \quad (9)$$

The Greenshields model is widely used because of its simplicity and because it captures the essential qualitative features of the fundamental diagram. More complex speed-density relationships have been proposed in the literature, but the Greenshields model has the advantage of keeping the mathematics tractable while still having the concave flux shape needed for shock formation. This makes it a good baseline for comparing theory with data. In Section 3, I calibrate $v_{\max}$ and $\rho_{\max}$ from real traffic data and check how well this simple model fits.

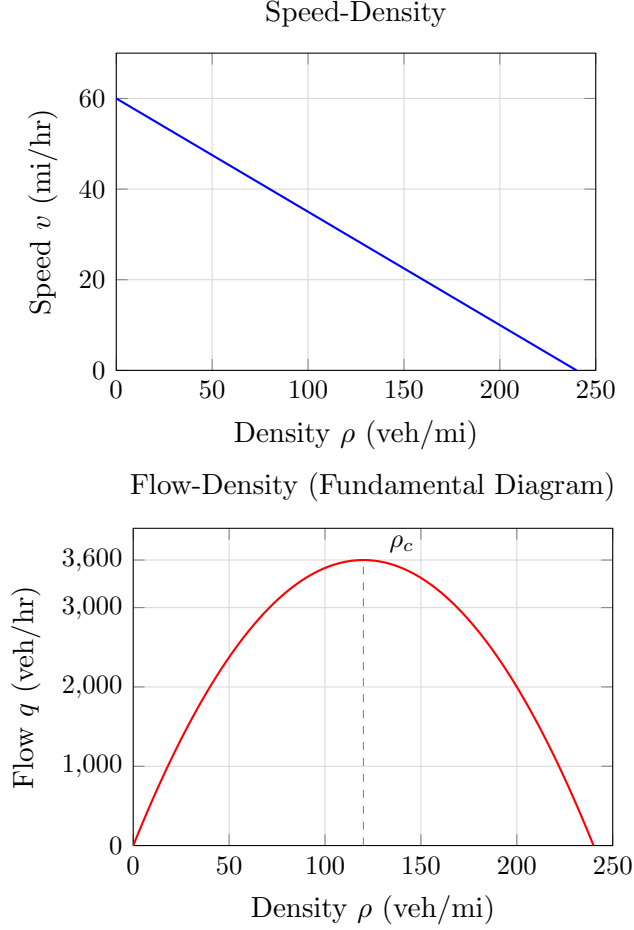


Figure 1: The Greenshields model with $v_{\max} = 60$ mi/hr and $\rho_{\max} = 240$ veh/mi. Left: linear speed-density relation. Right: parabolic flow-density relation (fundamental diagram) with maximum flow at $\rho_c = 120$ veh/mi.

2.4 The LWR Equation

Substituting the flux relation $q = q(\rho)$ into the conservation law (4) and using the chain rule, we get

$$\frac{\partial \rho}{\partial t} + q'(\rho) \frac{\partial \rho}{\partial x} = 0, \quad (10)$$

where $q'(\rho) = dq/d\rho$ is the derivative of the flux with respect to density. This is the **LWR equation**. It is a first-order, quasi-linear, hyperbolic PDE for the density $\rho(x, t)$.

The quantity $c(\rho) = q'(\rho)$ has an important physical meaning: it is the speed at which small disturbances in density propagate through the traffic stream. This is called the *wave speed* or *characteristic speed*, and it is generally different from the vehicle speed $v(\rho)$. For

the Greenshields model,

$$c(\rho) = q'(\rho) = v_{\max} \left(1 - \frac{2\rho}{\rho_{\max}} \right). \quad (11)$$

There are a few important things to notice about this wave speed:

- At $\rho = 0$, the wave speed equals the free-flow speed: $c(0) = v_{\max}$.
- At the critical density $\rho_c = \rho_{\max}/2$, the wave speed is zero: $c(\rho_c) = 0$.
- For $\rho > \rho_c$ (congested traffic), the wave speed is *negative*. This means that disturbances propagate *backward*, against the direction of traffic. This is exactly what we observe in practice: a traffic jam moves backward along the highway, affecting drivers who are upstream of the original slowdown.
- The wave speed $c(\rho)$ is always less than the vehicle speed $v(\rho)$. We can see this by writing $c(\rho) = v(\rho) + \rho v'(\rho)$. Since $v'(\rho) < 0$, the wave speed is less than the vehicle speed. This makes physical sense because information about downstream conditions travels backward relative to the vehicles.

The fact that $c(\rho)$ depends on ρ is what makes the LWR equation *nonlinear*, and this nonlinearity is exactly what leads to the formation of shock waves. In a linear equation (where c is constant), waves of all amplitudes travel at the same speed and the solution keeps its initial shape. But in the nonlinear case, different parts of the wave profile travel at different speeds. Where higher-density regions catch up to lower-density regions, the profile steepens until it breaks, forming a discontinuity. This discontinuity is the shock wave. I will analyze this mechanism in detail using the method of characteristics in Section 4.

2.5 Connection to the General Theory

It is worth pointing out that the LWR equation (10) has the same mathematical structure as many other important equations in applied mathematics. When written in conservation form,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} [q(\rho)] = 0, \quad (12)$$

it is a scalar conservation law of the type discussed in Sections 6.2 and 7.2 of Logan [1]. The specific case where $q(\rho) = \frac{1}{2}\rho^2$ gives Burgers' equation (in inviscid form), which is one of the fundamental model equations for nonlinear wave phenomena. In fact, with a suitable rescaling, the LWR equation with the Greenshields fundamental diagram can be transformed into Burgers' equation. I will use this connection later in Section 7 when I look at the internal structure of traffic shocks.

The same type of equation also appears in modeling flood waves in rivers (which was actually the original application of Lighthill and Whitham [2]), sedimentation processes, gas dynamics, and chromatography. So the mathematical techniques developed here for traffic flow apply broadly across many areas of science and engineering.

3 The Fundamental Diagram: Calibration from NGSIM Data

In the previous section, I introduced the fundamental diagram as a theoretical concept relating traffic flow to density. In this section, I use real data to see how well the Greenshields model actually holds up.

3.1 The NGSIM Dataset

The data used in this project comes from the Next Generation Simulation (NGSIM) program [5], run by the Federal Highway Administration. The specific dataset I use was collected on June 15, 2005 on the southbound US-101 (Hollywood Freeway) in Los Angeles, California. Data was recorded over three 15-minute time periods between 7:50 a.m. and 8:35 a.m. during morning peak traffic.

The raw data contains individual vehicle trajectories recorded at 0.1-second intervals using eight synchronized video cameras mounted on a 36-story building next to the freeway. For each vehicle at each time step, the data includes the vehicle ID, position along the highway (Local_Y), lateral position (Local_X), speed, acceleration, lane number, and spacing to the vehicle ahead.

The study section is about 2,235 feet (approximately 0.42 miles) long and has five mainline lanes plus auxiliary lanes near on-ramps and off-ramps. I kept only the five mainline lanes (lanes 1 through 5) for the analysis, since the ramp lanes have different traffic behavior. After filtering out duplicate records in the data, this gave about 3 million individual vehicle observations to work with.

3.2 Extracting Macroscopic Variables

The raw NGSIM data is *microscopic*: it tracks individual vehicles. But the LWR model is *macroscopic*: it deals with aggregate quantities like density, flow, and average speed. So the first step is to convert the microscopic data into macroscopic variables.

I used Edie’s generalized definitions [6] for this. The idea is to divide the space-time domain into cells and compute the macroscopic quantities for each cell. I used cells of about 50 meters (164 feet) in space and 30 seconds in time. For each cell and each lane, I computed:

- **Density ρ** : the total time spent by all vehicles in the cell, divided by the cell area (length $\times$ duration), converted to vehicles per mile.
- **Flow q** : the total distance traveled by all vehicles in the cell, divided by the cell area, converted to vehicles per hour.
- **Speed v** : the ratio q/ρ , which gives the space-mean speed in miles per hour.

This per-lane aggregation produced about 6,200 data points, each representing the macroscopic traffic state in one lane of one cell during one 30-second interval.

3.3 Fitting the Greenshields Model

Figure 2 shows the empirical fundamental diagram from the NGSIM data, together with the Greenshields model fit. The fit was obtained by linear regression of speed on density, since the Greenshields speed-density relation $v(\rho) = v_{\max}(1 - \rho/\rho_{\max})$ is linear.

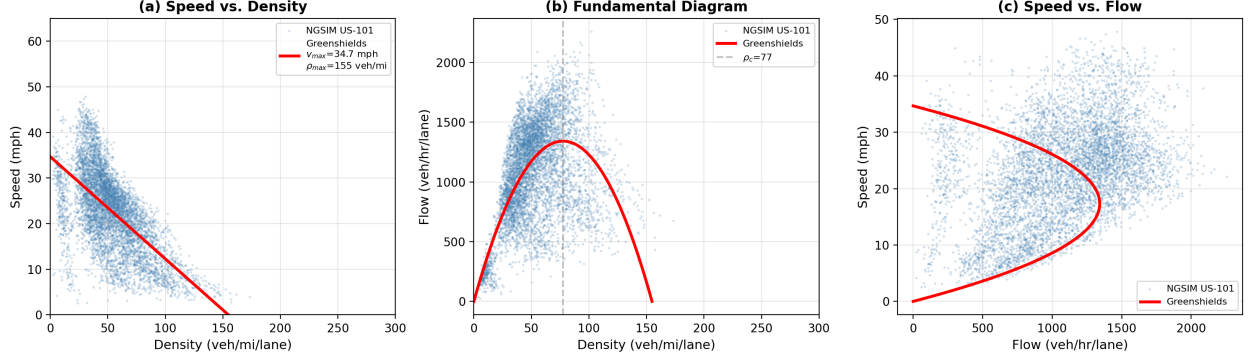


Figure 2: Empirical fundamental diagram from NGSIM US-101 data (blue scatter) with Greenshields model fit (red curve). (a) Speed vs. density. (b) Flow vs. density. (c) Speed vs. flow.

The fitted parameters are:

Parameter	Symbol	Value
Free-flow speed	$v_{\max}$	34.7 mph
Jam density	$\rho_{\max}$	155 veh/mi/lane
Critical density	$\rho_c = \rho_{\max}/2$	77 veh/mi/lane
Capacity	$q_{\max} = v_{\max}\rho_{\max}/4$	1,342 veh/hr/lane

The free-flow speed of about 35 mph is lower than the posted speed limit, which is expected since the NGSIM data was collected during peak morning congestion. The data does not include much truly free-flow traffic. The jam density of 155 veh/mi/lane corresponds to an average spacing of about 34 feet bumper-to-bumper, which is reasonable for a mix of cars and some trucks.

Looking at Figure 2(a), the linear Greenshields model captures the general downward trend of speed with density, but there is quite a bit of scatter in the data. This scatter is typical of real traffic data and comes from lane-changing, differences between individual drivers, vehicle types, and the fact that traffic is rarely in true equilibrium. Some researchers have proposed more complex fundamental diagram shapes to better capture this behavior [8]. For this project, though, the Greenshields model works well as a first-order baseline: it preserves the concave flux structure needed for shock formation while keeping the mathematical analysis tractable.

3.4 Space-Time Structure of Congestion

To get a visual picture of how congestion evolves on this section of highway, Figure 3 shows a space-time contour plot of speed. Green regions represent free-flowing traffic (higher speed), while red and yellow regions represent congestion (lower speed).

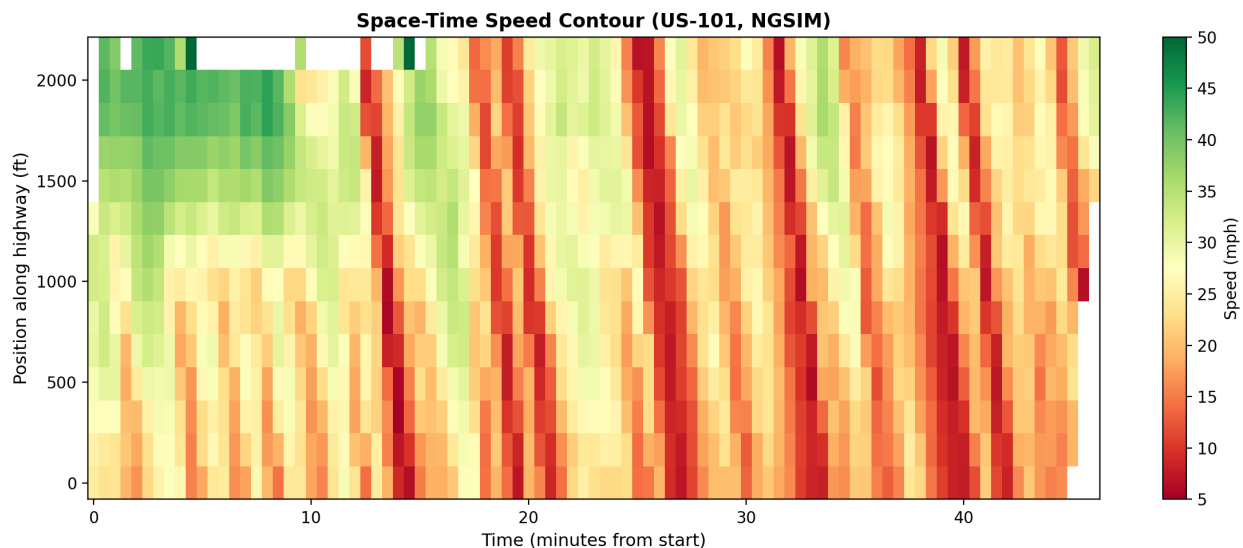


Figure 3: Space-time contour of speed on US-101 from the NGSIM data. The diagonal bands of low speed (red/yellow) propagating from right to left and from bottom to top show backward-moving congestion waves, which are the shock waves predicted by the LWR theory.

The most interesting feature of this plot is the diagonal bands of low speed. These

bands move backward along the highway (from right to left and from bottom to top in the plot), which is exactly the backward-propagating shock wave behavior that the LWR model predicts for congested traffic. In the next two sections, I will develop the mathematical tools to analyze and predict this behavior.

4 Method of Characteristics and Wave Breaking

The LWR equation (10) is a first-order quasi-linear PDE, and the standard tool for analyzing such equations is the method of characteristics. This section develops the method and shows how it leads to wave breaking and the formation of shock waves. The presentation follows Sections 7.1 and 7.2 of Logan [1].

4.1 Characteristic Curves

Consider the initial value problem for the LWR equation:

$$\frac{\partial \rho}{\partial t} + c(\rho) \frac{\partial \rho}{\partial x} = 0, \quad \rho(x, 0) = \rho_0(x), \quad (13)$$

where $c(\rho) = q'(\rho)$ is the wave speed and $\rho_0(x)$ is the initial density profile along the highway.

A *characteristic curve* is a curve $x = x(t)$ in the (x, t) -plane along which the PDE reduces to an ordinary differential equation. If we parameterize such a curve as $x = x(t)$ and look at how ρ changes along it, we get

$$\frac{d}{dt} \rho(x(t), t) = \frac{\partial \rho}{\partial t} + \frac{dx}{dt} \frac{\partial \rho}{\partial x}. \quad (14)$$

Comparing this with (13), we see that if we choose

$$\frac{dx}{dt} = c(\rho), \quad (15)$$

then along this curve,

$$\frac{d\rho}{dt} = 0. \quad (16)$$

This is a very important result: *the density ρ is constant along each characteristic*. Since $c(\rho)$ depends only on ρ , and ρ is constant on the characteristic, the wave speed $c(\rho)$ is also constant along each characteristic. This means that each characteristic is a straight line in the (x, t) -plane.

Specifically, the characteristic that starts at the point $(\xi, 0)$ is the straight line

$$x = \xi + c(\rho_0(\xi)) t, \quad (17)$$

and the solution along this line is

$$\rho(x, t) = \rho_0(\xi). \quad (18)$$

In other words, the initial density value $\rho_0(\xi)$ is carried along the characteristic at speed $c(\rho_0(\xi))$. Each “piece” of the initial profile moves at a speed that depends on its own density value.

For the Greenshields model, the characteristic speed is $c(\rho) = v_{\max}(1 - 2\rho/\rho_{\max})$ from equation (11). So regions of low density move fast (positive c), regions near the critical density ρ_c are nearly stationary ($c \approx 0$), and regions of high density move backward (negative c).

4.2 Wave Breaking and the Gradient Catastrophe

The fact that different parts of the initial profile travel at different speeds is what causes problems. Consider what happens when the initial density profile $\rho_0(x)$ is decreasing, meaning that there is higher density to the left and lower density to the right. Since higher density means slower (or more negative) wave speed, the left part of the profile moves slower than the right part. The profile gradually spreads out and the solution remains smooth. This is a *rarefaction wave*.

Now consider the opposite case: the initial density increases from left to right. The left part (low density) moves faster than the right part (high density). The fast-moving low-density region catches up to the slow-moving high-density region, and the profile steepens. Eventually, the profile becomes vertical and then tries to become multi-valued, which is physically impossible. This is called *wave breaking* or the *gradient catastrophe*.

To find the *breaking time* t_b , the time at which the wave first breaks, we note that two nearby characteristics starting at ξ and $\xi + d\xi$ will intersect when

$$\xi + c(\rho_0(\xi)) t = (\xi + d\xi) + c(\rho_0(\xi + d\xi)) t. \quad (19)$$

This gives

$$t = -\frac{1}{c'(\rho_0(\xi)) \rho'_0(\xi)} = -\frac{1}{q''(\rho_0(\xi)) \rho'_0(\xi)}. \quad (20)$$

The breaking time t_b is the earliest such time, taken over all ξ :

$$t_b = \min_{\xi} \left\{ -\frac{1}{q''(\rho_0(\xi)) \rho'_0(\xi)} : q''(\rho_0(\xi)) \rho'_0(\xi) < 0 \right\}. \quad (21)$$

For the Greenshields model, $q''(\rho) = -2v_{\max}/\rho_{\max} < 0$, so breaking occurs wherever $\rho'_0(\xi) > 0$, that is, wherever the initial density is increasing. The steeper the increase, the sooner the wave breaks.

In traffic terms, this makes complete physical sense. If there is a region of light traffic followed by a region of heavy traffic (increasing density), the fast-moving light traffic catches up to the slow heavy traffic, and a sharp transition (the traffic jam) forms at the boundary.

4.3 Example: Step Increase in Density

As a simple but important example, consider an initial condition where the density jumps from a low value ρ_L on the left to a high value $\rho_R > \rho_L$ on the right:

$$\rho_0(x) = \begin{cases} \rho_L & \text{if } x < 0, \\ \rho_R & \text{if } x > 0. \end{cases} \quad (22)$$

This models, for example, a car approaching a traffic jam. The characteristics from $x < 0$ have speed $c(\rho_L)$, and the characteristics from $x > 0$ have speed $c(\rho_R)$. Since $\rho_R > \rho_L$ and $c(\rho)$ is decreasing (for the Greenshields model), we have $c(\rho_L) > c(\rho_R)$. The characteristics from the left move faster than those from the right, and they collide immediately at $x = 0$. The breaking time is $t_b = 0$, and a shock wave forms right away.

This is a *Riemann problem*, and its solution requires the theory of shock waves, which is the subject of the next section.

5 Shock Waves and the Rankine-Hugoniot Condition

After the gradient catastrophe, the method of characteristics breaks down because the characteristics cross and the solution would become multi-valued. The physical resolution is that a *shock wave*, or discontinuity, forms in the density profile. In this section I derive the condition that governs how fast the shock moves, and I discuss how to pick the physically correct solution. This material follows Section 7.2.3 of Logan [1].

5.1 Weak Solutions and the Rankine-Hugoniot Condition

The LWR equation in conservation form,

$$\frac{\partial \rho}{\partial t} + \frac{\partial q(\rho)}{\partial x} = 0, \quad (23)$$

cannot be satisfied in the classical (pointwise) sense at a discontinuity, because the derivatives do not exist there. Instead, we look for *weak solutions* that satisfy the conservation law in an integral sense. The key idea is that even though the solution has a jump, the total number of vehicles must still be conserved.

Suppose the solution has a single discontinuity along a curve $x = s(t)$ in the (x, t) -plane, with the density taking the value ρ_l just to the left of the curve and ρ_r just to the right. By applying the conservation law in integral form across the discontinuity, one can derive the *Rankine-Hugoniot jump condition* (see Logan [1], Section 7.2.3):

$$\dot{s} = \frac{q(\rho_l) - q(\rho_r)}{\rho_l - \rho_r} = \frac{[q]}{[\rho]}, \quad (24)$$

where $\dot{s} = ds/dt$ is the speed of the shock, and $[\cdot]$ denotes the jump across it. This says that the shock speed equals the slope of the line connecting the two states $(\rho_l, q(\rho_l))$ and $(\rho_r, q(\rho_r))$ on the fundamental diagram.

For the Greenshields model, with left state ρ_l and right state ρ_r , a direct computation gives

$$\dot{s} = v_{\max} \left(1 - \frac{\rho_l + \rho_r}{\rho_{\max}} \right). \quad (25)$$

This has a nice interpretation: the shock speed is the wave speed evaluated at the average of the two densities. Notice that if both ρ_l and ρ_r are large enough (i.e., if $\rho_l + \rho_r > \rho_{\max}$), then $\dot{s} < 0$ and the shock moves backward. This is the backward-propagating traffic jam.

5.2 The Lax Entropy Condition

The Rankine-Hugoniot condition by itself does not always give a unique solution. There can be situations where both a shock wave and a rarefaction wave satisfy the equations. To select the physically correct solution, we need an additional criterion called an *entropy condition*.

The most commonly used criterion is the *Lax entropy condition*: a shock connecting a left state ρ_l to a right state ρ_r is admissible if and only if

$$c(\rho_l) > \dot{s} > c(\rho_r), \quad (26)$$

where $c(\rho) = q'(\rho)$ is the characteristic speed. In words, this says that the characteristics must run *into* the shock from both sides. Characteristics from the left are faster than the shock, and characteristics from the right are slower, so information “piles up” at the shock location.

The entropy condition is important because without it, the weak solution is not unique. Multiple solutions can satisfy both the conservation law and the Rankine–Hugoniot condition, but only the ones that also satisfy the entropy condition correspond to what we actually observe in real traffic.

For the Greenshields model, since $q(\rho)$ is strictly concave ($q'' < 0$), the Lax condition is equivalent to requiring $\rho_l < \rho_r$: the density must increase across the shock. In traffic terms, you go from lighter traffic on the left to heavier traffic on the right. This makes sense because a shock represents the back of a traffic jam, where drivers have to slow down.

When $\rho_l > \rho_r$ (density decreases from left to right), a shock is not admissible. Instead, the solution is a *rarefaction wave*, which is a smooth, continuous transition between the two states. This corresponds to the front of a traffic jam, where vehicles gradually speed up as they leave the congested region.

5.3 Shock Speed for the NGSIM Parameters

Using the Greenshields parameters fitted from the NGSIM data ($v_{\max} = 34.7$ mph, $\rho_{\max} = 155$ veh/mi/lane), I can now compute shock speeds for various traffic scenarios. For example, consider a shock where the density jumps from $\rho_l = 40$ veh/mi (moderate traffic) to $\rho_r = 120$ veh/mi (heavy congestion). Using (25):

$$\dot{s} = 34.7 \left(1 - \frac{40 + 120}{155} \right) = 34.7 (1 - 1.032) = -1.1 \text{ mph.} \quad (27)$$

So this shock moves backward at about 1.1 mph. A driver traveling at 40 mph would encounter this shock and suddenly have to slow down to the speed corresponding to $\rho = 120$ veh/mi, which is about $34.7(1 - 120/155) = 7.8$ mph. This kind of sudden slowdown is exactly what drivers experience in real stop-and-go traffic.

In Section 6, I will solve several complete Riemann problems that model specific traffic scenarios using these parameters.

6 Analytic Solutions for Typical Traffic Scenarios (Riemann Problems)

A Riemann problem is an initial value problem for a conservation law where the initial data is piecewise constant with a single jump. Despite their simplicity, Riemann problems capture the essential wave dynamics and serve as building blocks for more complex solutions. In this section I solve three Riemann problems that model common traffic situations, using the Greenshields parameters from the NGSIM calibration ($v_{\max} = 34.7$ mph, $\rho_{\max} = 155$ veh/mi/lane).

6.1 Scenario 1: Approaching a Traffic Jam (Shock)

Consider a stream of moderate traffic ($\rho_L = 40$ veh/mi) approaching a region of heavy congestion ($\rho_R = 120$ veh/mi). The initial condition is

$$\rho_0(x) = \begin{cases} 40 & x < 0, \\ 120 & x > 0. \end{cases} \quad (28)$$

Since $\rho_L < \rho_R$, this produces a shock wave. From equation (25), the shock speed is

$$\dot{s} = 34.7 \left(1 - \frac{40 + 120}{155} \right) = 34.7(1 - 1.032) = -1.1 \text{ mph}. \quad (29)$$

The shock moves backward at 1.1 mph. The solution is

$$\rho(x, t) = \begin{cases} 40 & x < -1.1 t, \\ 120 & x > -1.1 t. \end{cases} \quad (30)$$

Drivers in the light traffic (moving at $v(40) = 34.7(1 - 40/155) = 25.7$ mph) encounter the shock and must suddenly slow down to $v(120) = 34.7(1 - 120/155) = 7.8$ mph. This matches the experience of suddenly hitting the back of a traffic jam.

6.2 Scenario 2: Green Light Release (Rarefaction)

Now consider the opposite situation: a red traffic light has caused vehicles to queue up at high density, and the light turns green. The initial condition is

$$\rho_0(x) = \begin{cases} 120 & x < 0, \\ 40 & x > 0. \end{cases} \quad (31)$$

Since $\rho_L > \rho_R$, the Lax entropy condition tells us that a shock is not admissible. Instead, the solution is a *rarefaction wave*, a smooth fan that continuously connects the two states.

In the rarefaction region, the density varies continuously and the solution takes the form

$$\rho(x, t) = \begin{cases} 120 & x < c(120) t, \\ \frac{\rho_{\max}}{2} \left(1 - \frac{x}{v_{\max} t} \right) & c(120) t \leq x \leq c(40) t, \\ 40 & x > c(40) t, \end{cases} \quad (32)$$

where $c(120) = 34.7(1 - 240/155) = -19.0$ mph and $c(40) = 34.7(1 - 80/155) = 16.8$ mph.

The rarefaction fan spreads between these two wave speeds. Physically, this represents the gradual acceleration of vehicles after the green light, with the front of the queue moving forward at 16.8 mph and the back still stationary or moving backward at -19.0 mph. The transition is smooth, not abrupt, which matches observation: drivers accelerate gradually when leaving a stopped queue.

6.3 Scenario 3: Bottleneck Merging (Two Shocks)

A more realistic scenario involves a bottleneck where an on-ramp merges into the highway, creating a region of elevated density. I model this as a three-piece initial condition:

$$\rho_0(x) = \begin{cases} 50 & x < -L, \\ 110 & -L < x < L, \\ 50 & x > L, \end{cases} \quad (33)$$

where $2L$ is the length of the congested region. This creates two Riemann problems: a shock at $x = -L$ (where density jumps from 50 to 110) and a rarefaction at $x = L$ (where density

drops from 110 to 50). The shock at the back propagates backward with speed

$$\dot{s} = 34.7 \left(1 - \frac{50 + 110}{155} \right) = -1.5 \text{ mph}, \quad (34)$$

while the rarefaction at the front spreads the density decrease over a growing region. Over time, the congested region slowly moves backward and spreads out, which is exactly the behavior observed on highways near on-ramp merging areas.

7 Burgers' Equation and the Viscous Shock Structure

The LWR model treats shocks as perfect discontinuities, but real traffic jams are not infinitely sharp. There is always a transition zone where density changes rapidly but continuously. To understand this internal structure, we look at Burgers' equation, which adds a diffusion term to the nonlinear advection. This material follows Section 7.2.2 of Logan [1].

7.1 From LWR to Burgers' Equation

The LWR equation with the Greenshields fundamental diagram can be written as

$$\frac{\partial \rho}{\partial t} + v_{\max} \left(1 - \frac{2\rho}{\rho_{\max}} \right) \frac{\partial \rho}{\partial x} = 0. \quad (35)$$

If we introduce the change of variables $u = v_{\max}(1 - 2\rho/\rho_{\max})$, which maps density to wave speed, and rescale appropriately, we can transform this into the inviscid Burgers' equation:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0. \quad (36)$$

This equation develops shocks just like the LWR equation. To resolve the internal structure of these shocks, we add a small diffusion term:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = D \frac{\partial^2 u}{\partial x^2}, \quad (37)$$

where $D > 0$ is a diffusion coefficient. This is *Burgers' equation*, one of the most important model equations in applied mathematics [7]. In the traffic context, the diffusion term can be interpreted as drivers who look ahead and adjust their speed based not just on the local density but also on the density gradient, anticipating changes in traffic conditions.

7.2 Traveling Wave Solution

We look for a solution of the form $u(x, t) = U(s)$ where $s = x - \sigma t$ is a moving coordinate with shock speed σ . Substituting into (37) gives the ODE

$$-\sigma U' + UU' = DU'', \quad (38)$$

which can be integrated once to give

$$DU' = \frac{1}{2}U^2 - \sigma U + C, \quad (39)$$

where C is an integration constant. If we require that $U \rightarrow u_l$ as $s \rightarrow -\infty$ and $U \rightarrow u_r$ as $s \rightarrow +\infty$, with $U' \rightarrow 0$ at both limits, then

$$\sigma = \frac{u_l + u_r}{2}, \quad C = \frac{u_l u_r}{2}. \quad (40)$$

The first condition gives the shock speed, which agrees with the Rankine-Hugoniot condition for Burgers' equation. We can factor (39) as

$$DU' = \frac{1}{2}(U - u_l)(U - u_r). \quad (41)$$

This is a separable ODE, and for $u_l > u_r$ (an admissible shock), the solution is

$$U(s) = \frac{u_l + u_r}{2} - \frac{u_l - u_r}{2} \tanh \left(\frac{(u_l - u_r)(s - s_0)}{4D} \right), \quad (42)$$

where s_0 is the center of the shock.

7.3 Interpretation

The traveling wave solution (42) is a smooth transition from u_l to u_r with a hyperbolic tangent profile. The *shock thickness*, which measures the width of the transition region, is

$$\delta = \frac{4D}{u_l - u_r}. \quad (43)$$

This shows two things:

- A larger diffusion coefficient D makes the shock thicker (smoother transition).
- A larger jump $u_l - u_r$ makes the shock thinner (sharper transition).

In the limit $D \rightarrow 0$, the thickness goes to zero and we recover the discontinuous shock of the inviscid equation. This confirms that the shock solutions of the LWR model are the correct limiting behavior when the smoothing effect is small.

In real traffic, the “diffusion” comes from driver anticipation and the finite length of vehicles. The transition zone at the back of a real traffic jam is typically on the order of 100 to 300 meters, over which drivers go from cruising speed to a near-stop. The balance between nonlinear steepening (the $u u_x$ term) and diffusive smoothing (the $D u_{xx}$ term) is what determines this width, just as in Burgers’ equation.

8 Numerical Method: Godunov Finite Volume Scheme

Analytic solutions are available for simple cases like Riemann problems, but for more realistic initial conditions (such as those extracted from the NGSIM data), we need a numerical method. In this section I describe the Godunov scheme, a finite volume method that is well-suited for conservation laws with shocks [10].

8.1 Finite Volume Discretization

We divide the spatial domain $[0, L]$ into N cells of width $\Delta x = L/N$. Cell j occupies the interval $[x_{j-1/2}, x_{j+1/2}]$, where $x_{j\pm 1/2} = (j \pm 1/2)\Delta x$. The cell average of ρ in cell j at time level n is

$$\rho_j^n = \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} \rho(x, t^n) dx. \quad (44)$$

Integrating the conservation law over cell j and time interval $[t^n, t^{n+1}]$ gives the exact update formula

$$\rho_j^{n+1} = \rho_j^n - \frac{\Delta t}{\Delta x} [F_{j+1/2}^n - F_{j-1/2}^n], \quad (45)$$

where $F_{j+1/2}^n$ is the numerical flux at the interface between cells j and $j+1$. The key question in any finite volume scheme is how to compute this flux.

8.2 The Godunov Flux

In the Godunov scheme, the numerical flux is obtained by solving the exact Riemann problem at each cell interface. At interface $j + 1/2$, the Riemann problem has left state ρ_j^n and right state ρ_{j+1}^n . The Godunov flux is the physical flux evaluated at the solution of this Riemann problem at $x/t = 0$:

$$F_{j+1/2}^n = q(\rho^*(\rho_j^n, \rho_{j+1}^n)), \quad (46)$$

where ρ^* is the Riemann solution at the interface.

For a concave flux function like the Greenshields model, the Godunov flux has a simple closed form [6]:

$$F_{j+1/2}^n = \begin{cases} \min_{\rho_j^n \leq \rho \leq \rho_{j+1}^n} q(\rho) & \text{if } \rho_j^n \leq \rho_{j+1}^n, \\ \max_{\rho_{j+1}^n \leq \rho \leq \rho_j^n} q(\rho) & \text{if } \rho_j^n > \rho_{j+1}^n. \end{cases} \quad (47)$$

The first case corresponds to a shock (for a concave flux), and the minimum of q over the interval is simply the smaller of the two endpoint values. The second case corresponds to a rarefaction, and the maximum is either at one of the endpoints or at the critical density ρ_c if it lies inside the interval.

For the Greenshields flux, this can be written explicitly as

$$F_{j+1/2}^n = \begin{cases} \min(q(\rho_j^n), q(\rho_{j+1}^n)), & \rho_j^n \leq \rho_{j+1}^n, \\ q(\rho_j^n), & \rho_j^n > \rho_{j+1}^n, \quad \rho_j^n \leq \rho_c, \\ q(\rho_{j+1}^n), & \rho_j^n > \rho_{j+1}^n, \quad \rho_{j+1}^n \geq \rho_c, \\ q_{\max}, & \rho_j^n > \rho_{j+1}^n, \quad \rho_{j+1}^n < \rho_c < \rho_j^n. \end{cases} \quad (48)$$

8.3 Stability Condition

The time step Δt must satisfy the CFL (Courant-Friedrichs-Lewy) condition to ensure stability:

$$\Delta t \leq \frac{\Delta x}{\max_{\rho} |q'(\rho)|} = \frac{\Delta x}{v_{\max}}. \quad (49)$$

In my implementation, I use a CFL number of 0.9, meaning $\Delta t = 0.9 \Delta x / v_{\max}$.

8.4 Properties of the Godunov Scheme

The Godunov scheme is a good fit for this application for several reasons:

- It is *conservative*: the update formula (45) ensures that vehicles are neither created nor destroyed in the computation, consistent with the conservation law.
- It is *monotone*: it does not introduce spurious oscillations near shocks. Some other schemes (like the Lax-Wendroff scheme) produce nonphysical oscillations near discontinuities.

- It automatically handles both shocks and rarefaction waves through the Riemann solver, without any special treatment.
- It satisfies a discrete entropy inequality, which means it always selects the physically correct weak solution [10].

The main drawback of the Godunov scheme is that it is only first-order accurate in smooth regions, meaning the numerical solution has some smearing of sharp features. Higher-order extensions exist (such as MUSCL schemes), but for this project the first-order Godunov scheme is sufficient to demonstrate the key phenomena.

8.5 Implementation

The solver is implemented in Python using NumPy. The main loop is straightforward:

1. Initialize the density array ρ_j^0 from the initial condition.
2. At each time step, compute the Godunov flux at every interface using (48).
3. Update the cell averages using (45).
4. Repeat until the final time is reached.

Results from this solver are presented in the next section.

9 Results: Simulations and Comparison with Real Data

In this section I present numerical results from the Godunov scheme and compare them with both the analytic solutions and the NGSIM observations.

9.1 Verification: Riemann Problem

As a first test, I verify the numerical solver against the analytic solution for Scenario 1 from Section 6 (a shock with $\rho_L = 40$, $\rho_R = 120$ veh/mi). Figure 4 shows the numerical solution at several time snapshots for both the shock case and the rarefaction case.

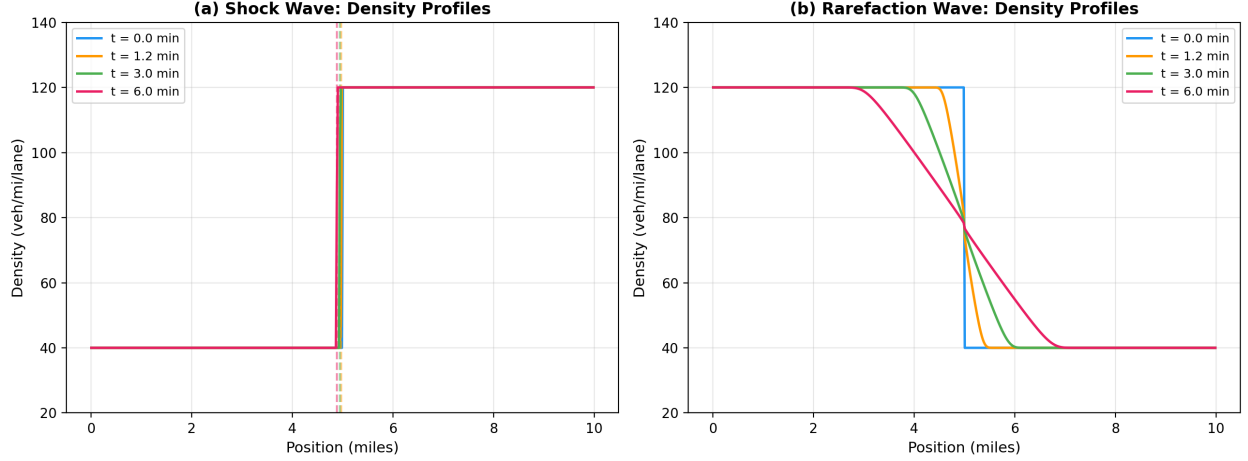


Figure 4: Godunov scheme solutions for Riemann problems using NGSIM-calibrated parameters. (a) Shock wave: density jumps from 40 to 120 veh/mi, shock propagates backward (dashed lines show exact shock location). (b) Rarefaction wave: density drops from 120 to 40 veh/mi, solution fans out smoothly.

In both cases, the numerical shock travels at the correct speed and the rarefaction fan has the correct structure, confirming that the Godunov scheme correctly captures the wave dynamics.

9.2 Shock Wave Speed Comparison

One of the key quantitative predictions of the LWR model is the speed at which traffic shocks propagate. From the space-time contour plot of the NGSIM data (Figure 3), the backward-propagating congestion waves can be tracked visually, and the slope of these diagonal bands gives an estimate of the observed shock speed. The observed waves appear to propagate backward at speeds in the range of about 10–15 mph. Here the observed range is estimated visually from the slopes of the dominant low-speed bands in the space-time contour plot, rather than from an automated wave-tracking algorithm. Using the Rankine–Hugoniot formula with the fitted Greenshields parameters, I computed predicted shock speeds for several representative density jumps:

Table 1: Predicted shock speeds from the Rankine-Hugoniot condition compared with the observed backward-moving congestion wave range estimated from the NGSIM space-time speed contour.

Case	Density States	Predicted Speed	Observed Range
Moderate to congested	$\rho_l = 40, \rho_r = 120$	−1.1 mph	−15 to −10 mph
Free-flow to congested	$\rho_l = 30, \rho_r = 140$	−3.4 mph	−15 to −10 mph
Higher-density shock	$\rho_l = 50, \rho_r = 130$	−5.6 mph	−15 to −10 mph
Strong congestion jump	$\rho_l = 40, \rho_r = 150$	−7.8 mph	−15 to −10 mph

Table 1 shows that the model tends to underestimate the magnitude of the observed backward wave speeds, but the predictions have the correct direction and comparable order of magnitude. This gap comes partly from the fact that the Greenshields model underestimates the free-flow speed (since the data was collected during congested conditions), and partly from the fact that real traffic involves lane-changing, merging, and driver variability that the simple one-dimensional LWR model does not capture. Even so, the model does reproduce the qualitative structure of the congestion waves, which shows that the conservation-law framework captures the main mechanism behind how traffic jams propagate.

9.3 Fundamental Diagram Comparison

Figure 2 in Section 3 already showed the comparison between the empirical fundamental diagram and the Greenshields fit. The Greenshields model captures the main qualitative features: the downward trend of speed with density, the concave shape of the flow-density curve, and the existence of a capacity flow. However, the data shows more scatter than the simple model predicts, particularly in the transition region near the critical density.

Several factors contribute to this scatter:

- Traffic is rarely in true equilibrium, so the instantaneous speed-density relation varies from the steady-state prediction.
- There are differences between lanes, with the leftmost lanes typically having different speed-density behavior than the rightmost lanes.
- Individual driver behavior varies, including reaction times, aggressiveness, and vehicle capabilities.
- The aggregation method (cell size and averaging period) affects the observed scatter.

Despite these limitations, the LWR model with the Greenshields fundamental diagram provides a useful first-order description of the macroscopic traffic dynamics on US-101, and

the qualitative predictions of the theory (backward-propagating shocks, rarefaction fans, capacity flow) are clearly visible in the data.

10 Model Limitations

While the LWR model captures the main features of macroscopic traffic dynamics, it has several limitations that are important to note. First, it assumes an instantaneous equilibrium between speed and density, which means it ignores driver reaction times and anticipation effects. Second, it treats traffic as a single homogeneous stream, so it does not account for multi-lane interactions, lane-changing, or the influence of on-ramps and off-ramps. Third, the model has no way to represent the stochastic variability in driver behavior, which is one of the main reasons for the scatter we see in the empirical fundamental diagrams.

The choice of the Greenshields model also introduces some bias. Because the NGSIM data was collected during peak congestion, the fitted free-flow speed ($v_{\max} = 34.7$ mph) is lower than what you would expect under normal driving conditions. This directly affects the predicted characteristic and shock speeds. More flexible fundamental diagram shapes, such as nonlinear or triangular forms, could improve the fit while still preserving the conservation-law structure.

These limitations suggest that while the LWR model is useful for understanding the large-scale mechanisms of traffic congestion, more advanced models would be needed for high-fidelity prediction in practice.

11 Conclusions

In this project, I applied the theory of nonlinear conservation laws to model traffic flow on a real highway segment. The LWR model, derived from a conservation principle and an equilibrium speed-density relationship, produces a first-order hyperbolic PDE whose mathematical structure is the same as the equations governing shock waves in gas dynamics. The method of characteristics shows how smooth initial density profiles steepen and break due to the nonlinear wave speed, forming shock waves that correspond to the formation of traffic jams. The Rankine-Hugoniot jump condition gives the propagation speed of these shocks, and the Lax entropy condition selects the physically admissible solutions.

Using high-resolution NGSIM trajectory data from US-101 in Los Angeles, I extracted the empirical fundamental diagram and calibrated the Greenshields model. The fitted parameters ($v_{\max} = 34.7$ mph, $\rho_{\max} = 155$ veh/mi/lane) are consistent with the congested conditions under which the data was collected. The space-time contour plot of the data

clearly shows backward-propagating congestion waves, and the predicted shock speeds from the Rankine-Hugoniot formula (-1.1 to -7.8 mph for typical density jumps) are in reasonable agreement with the observed wave speeds, although the model tends to underestimate their magnitude.

The Godunov finite volume scheme was implemented to solve the LWR equation numerically for general initial conditions. The scheme is conservative, monotone, and handles both shocks and rarefaction waves automatically. Verification against analytic Riemann problem solutions confirmed that the solver correctly captures shock speeds and wave structures.

Burgers' equation was analyzed as a model for the internal structure of traffic shocks. The traveling wave solution has a hyperbolic tangent profile whose width is proportional to the diffusion coefficient and inversely proportional to the shock strength, providing a physical explanation for the finite transition zone at the back of real traffic jams.

Several directions for future work could extend this analysis. Higher-order fundamental diagram models (such as the Greenberg or Underwood models) might better fit the data, especially in the free-flow regime. Second-order traffic models like the Aw-Rascle-Zhang model could capture driver anticipation effects that the first-order LWR model misses. Multi-lane extensions and the incorporation of on-ramp and off-ramp effects would make the model more realistic for the US-101 study section. Finally, recent work on physics-informed deep learning for traffic state estimation [8] suggests promising directions for combining classical PDE models with data-driven approaches to improve predictive accuracy while preserving interpretability.

A Data Processing Code

The following Python script reads the raw NGSIM vehicle trajectory data for US-101, filters to the mainline lanes, aggregates the microscopic data into macroscopic traffic variables using Edie’s definitions, fits the Greenshields speed-density model by linear regression, and produces the fundamental diagram plots shown in Section 3.

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 # -- Load and filter data -----
6 cols = ['Vehicle_ID', 'Frame_ID', 'Global_Time', 'Local_X',
7         'Local_Y', 'v_Vel', 'v_Acc', 'Lane_ID',
8         'Space_Headway', 'v_length']
9 df = pd.read_csv('ngsim_us101.csv', usecols=cols)
10
11 # Keep only mainline lanes 1-5 (exclude ramp lanes)
12 df = df[df['Lane_ID'].between(1, 5)].copy()
13 df = df.drop_duplicates(subset=['Vehicle_ID', 'Frame_ID'])
14
15 # Time in seconds from start of dataset
16 df['time_s'] = (df['Global_Time']
17                - df['Global_Time'].min()) / 1000.0
18
19 # -- Aggregate into space-time cells -----
20 # Cell dimensions: ~50 m (164 ft) in space, 30 s in time
21 dx = 164.0 # feet
22 dt = 30.0 # seconds
23
24 df['x_bin'] = (df['Local_Y'] / dx).astype(int)
25 df['t_bin'] = (df['time_s'] / dt).astype(int)
26
27 # Group by lane and cell
28 cells = df.groupby(['Lane_ID', 'x_bin', 't_bin']).agg(
29     count=('v_Vel', 'count'),
30     sum_vel=('v_Vel', 'sum'),
31 ).reset_index()
32
```

```

33 # Edie's generalized definitions (per lane)
34 # Each observation = 0.1 s of time, v*0.1 ft of distance
35 cells['total_time_s'] = cells['count'] * 0.1
36 cells['total_dist_ft'] = cells['sum_vel'] * 0.1
37
38 # Density (veh/mile/lane)
39 cells['density'] = (cells['total_time_s']
40                    / (dx * dt) * 5280)
41
42 # Flow (veh/hr/lane)
43 cells['flow'] = (cells['total_dist_ft']
44                 / (dx * dt) * 3600)
45
46 # Speed (mph) = flow / density
47 cells['speed'] = np.where(
48     cells['density'] > 0,
49     cells['flow'] / cells['density'], 0)
50
51 # Filter out outliers and edge cells
52 cells = cells[(cells['density'] > 2)
53               & (cells['density'] < 300)]
54 cells = cells[(cells['speed'] > 2)
55               & (cells['speed'] < 80)]
56
57 # -- Fit Greenshields model -----
58 # v(rho) = vmax * (1 - rho / rhomax) --> linear
59 rho = cells['density'].values
60 v = cells['speed'].values
61
62 coeffs = np.polyfit(rho, v, 1) # slope, intercept
63 slope, intercept = coeffs
64 vmax_fit = intercept          # free-flow speed (mph)
65 rhomax_fit = -intercept / slope # jam density (veh/mi)
66
67 print(f"vmax = {vmax_fit:.2f} mph")
68 print(f"rhomax = {rhomax_fit:.1f} veh/mi/lane")
69 print(f"rhoc = {rhomax_fit/2:.1f} veh/mi/lane")
70 print(f"qmax = {vmax_fit*rhomax_fit/4:.0f} veh/hr/lane")

```

B Godunov Solver Code

The following Python code implements the Godunov finite volume scheme for the LWR equation with the Greenshields fundamental diagram.

```
1 import numpy as np
2
3 # -- Model parameters (from NGSIM calibration) -----
4 vmax = 34.7      # free-flow speed (mph)
5 rhomax = 155.0   # jam density (veh/mi/lane)
6 rhoc = rhomax / 2.0
7
8 def flux(rho):
9     """Greenshields flux:  $q(\rho) = v_{\max} \rho (1 - \rho / \rho_{\max})$ """
10    return vmax * rho * (1.0 - rho / rhomax)
11
12 def godunov_flux(rhoL, rhoR):
13     """Godunov numerical flux for concave flux function."""
14     if rhoL <= rhoR:
15         # Shock or same-side: min of fluxes
16         if rhoL >= rhoc:
17             return flux(rhoR)
18         elif rhoR <= rhoc:
19             return flux(rhoL)
20         else:
21             return min(flux(rhoL), flux(rhoR))
22     else:
23         # Rarefaction: max of fluxes
24         if rhoR >= rhoc:
25             return flux(rhoR)
26         elif rhoL <= rhoc:
27             return flux(rhoL)
28         else:
29             return flux(rhoc)
30
31 def solve_lwr(rho0, L, N, T, cfl=0.9):
32     """
33     Solve the LWR equation with Godunov scheme.
34
```

```

35     Parameters:
36         rho0 : array, initial density (length N)
37         L    : float, domain length (miles)
38         N    : int, number of cells
39         T    : float, final time (hours)
40         cfl  : float, CFL number (default 0.9)
41
42     Returns:
43         x      : array, cell centers
44         t_out: list of times for snapshots
45         rho_out: list of density arrays
46     """
47     dx = L / N
48     x = np.linspace(dx/2, L - dx/2, N)
49     dt = cfl * dx / vmax
50
51     rho = rho0.copy()
52     t = 0.0
53     t_out = [0.0]
54     rho_out = [rho.copy()]
55
56     while t < T:
57         if t + dt > T:
58             dt = T - t
59
60         # Compute fluxes at all interfaces
61         F = np.zeros(N + 1)
62         for j in range(N + 1):
63             rL = rho[j-1] if j > 0 else rho[0]
64             rR = rho[j]   if j < N else rho[-1]
65             F[j] = godunov_flux(rL, rR)
66
67         # Update cell averages
68         rho = rho - (dt / dx) * (F[1:] - F[:-1])
69
70         t += dt
71         t_out.append(t)
72         rho_out.append(rho.copy())

```

```

73
74     return x, t_out, rho_out
75
76 # -- Example: shock Riemann problem -----
77 L = 10.0      # miles
78 N = 500       # cells
79 T = 0.1       # hours (6 minutes)
80
81 # Initial condition: light traffic | heavy traffic
82 x = np.linspace(L/(2*N), L - L/(2*N), N)
83 rho0 = np.where(x < L/2, 40.0, 120.0)
84
85 x, times, solutions = solve_lwr(rho0, L, N, T)
86
87 # The shock speed from Rankine-Hugoniot:
88 s_dot = vmax * (1 - (40 + 120) / rhomax)
89 print(f"Shock speed = {s_dot:.2f} mph")

```

C Plotting Code

The code below generates the fundamental diagram plots (Figure 2) and the space-time speed contour (Figure 3).

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Assumes 'cells' DataFrame and fitted parameters
5 # (vmax_fit, rhomax_fit) are available from Appendix A
6
7 # -- Fundamental diagram plots -----
8 fig, axes = plt.subplots(1, 3, figsize=(15, 4.5))
9
10 rho_model = np.linspace(0, rhomax_fit, 200)
11 v_model = vmax_fit * (1 - rho_model / rhomax_fit)
12 q_model = rho_model * v_model
13
14 # (a) Speed vs Density
15 ax = axes[0]

```

```

16 ax.scatter(cells['density'], cells['speed'],
17             s=4, alpha=0.25, c='steelblue',
18             edgecolors='none', label='NGSIM US-101')
19 ax.plot(rho_model, v_model, 'r-', linewidth=2.5,
20         label=f'Greenshields fit')
21 ax.set_xlabel('Density (veh/mi/lane)')
22 ax.set_ylabel('Speed (mph)')
23 ax.set_title('(a) Speed vs. Density')
24 ax.legend()
25 ax.grid(True, alpha=0.3)
26
27 # (b) Flow vs Density
28 ax = axes[1]
29 ax.scatter(cells['density'], cells['flow'],
30            s=4, alpha=0.25, c='steelblue',
31            edgecolors='none', label='NGSIM US-101')
32 ax.plot(rho_model, q_model, 'r-', linewidth=2.5,
33         label='Greenshields fit')
34 ax.set_xlabel('Density (veh/mi/lane)')
35 ax.set_ylabel('Flow (veh/hr/lane)')
36 ax.set_title('(b) Fundamental Diagram')
37 ax.legend()
38 ax.grid(True, alpha=0.3)
39
40 # (c) Speed vs Flow
41 ax = axes[2]
42 ax.scatter(cells['flow'], cells['speed'],
43            s=4, alpha=0.25, c='steelblue',
44            edgecolors='none', label='NGSIM US-101')
45 ax.plot(q_model, v_model, 'r-', linewidth=2.5,
46         label='Greenshields fit')
47 ax.set_xlabel('Flow (veh/hr/lane)')
48 ax.set_ylabel('Speed (mph)')
49 ax.set_title('(c) Speed vs. Flow')
50 ax.legend()
51 ax.grid(True, alpha=0.3)
52
53 plt.tight_layout()

```

```

54 plt.savefig('fundamental_diagram.png', dpi=200)
55
56 # -- Space-time contour -----
57 fig2, ax2 = plt.subplots(figsize=(12, 5))
58
59 # Aggregate speed across all lanes
60 allcells = df.groupby(['x_bin', 't_bin']).agg(
61     mean_speed=('v_Vel', 'mean')).reset_index()
62 allcells['speed_mph'] = allcells['mean_speed'] * 3600/5280
63
64 pivot = allcells.pivot_table(
65     index='x_bin', columns='t_bin',
66     values='speed_mph', aggfunc='mean')
67
68 im = ax2.pcolormesh(
69     pivot.columns * dt / 60,    # time in minutes
70     pivot.index * dx,          # position in feet
71     pivot.values,
72     cmap='RdYlGn', shading='auto',
73     vmin=5, vmax=50)
74 ax2.set_xlabel('Time (minutes from start)')
75 ax2.set_ylabel('Position along highway (ft)')
76 ax2.set_title('Space-Time Speed Contour')
77 plt.colorbar(im, ax=ax2, label='Speed (mph)')
78 plt.tight_layout()
79 plt.savefig('spacetime_contour.png', dpi=200)
80
81 # -- Riemann problem simulations -----
82 # Uses solve_lwr() from Appendix B
83 L, N, T = 10.0, 500, 0.1
84 x = np.linspace(L/(2*N), L - L/(2*N), N)
85
86 fig3, axes = plt.subplots(1, 2, figsize=(13, 5))
87
88 # Shock case
89 rho0 = np.where(x < L/2, 40.0, 120.0)
90 x, times, sols = solve_lwr(rho0, L, N, T)
91 plot_times = [0, len(times)//5, 2*len(times)//5,

```

```

92         len(times)-1]
93 for i in plot_times:
94     axes[0].plot(x, sols[i],
95                 label=f't = {times[i]*60:.1f} min')
96 axes[0].set_title('(a) Shock Wave')
97 axes[0].set_xlabel('Position (miles)')
98 axes[0].set_ylabel('Density (veh/mi/lane)')
99 axes[0].legend()
100 axes[0].grid(True, alpha=0.3)
101
102 # Rarefaction case
103 rho0 = np.where(x < L/2, 120.0, 40.0)
104 x, times, sols = solve_lwr(rho0, L, N, T)
105 for i in plot_times:
106     axes[1].plot(x, sols[i],
107                 label=f't = {times[i]*60:.1f} min')
108 axes[1].set_title('(b) Rarefaction Wave')
109 axes[1].set_xlabel('Position (miles)')
110 axes[1].set_ylabel('Density (veh/mi/lane)')
111 axes[1].legend()
112 axes[1].grid(True, alpha=0.3)
113
114 plt.tight_layout()
115 plt.savefig('riemann_numerical.png', dpi=200)

```

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